

Notes: Banerjee, Breza, Chandrasekhar, Duflo, Jackson and Kinnan (RES, 2024)

Changes in Social Network Structure in Response to Exposure to Formal Credit Markets

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Two contexts	4
2.2	Network measurement	4
2.3	Classifying households as likely or unlikely microfinance borrowers	5
3	Empirical specifications and model (all equations + interpretation)	5
3.1	Graph-level difference-in-differences in Karnataka	5
3.2	Cross-sectional treatment effect in Hyderabad	6
3.3	Link-level evolution by household type	6
3.4	Triangle-level evolution	6
3.5	Risk-sharing specification	7
3.6	Network-formation model: undirected socializing effort	7
3.7	Specialization to microfinance	8
4	Identification and instruments (what makes the estimates credible)	8
4.1	Karnataka: difference-in-differences	8
4.2	Hyderabad: randomized early entry	9
4.3	Why the two settings complement each other	9
5	Tables and extracted visuals (original numbers and titles)	10
5.1	TABLE 1. Sample statistics and TABLE 1. Continued	10
5.2	TABLE 2. Characteristics of H versus L and TABLE 3. Effects of MF on graph-level characteristics	11
5.3	TABLE 4. Link evolution, Karnataka and TABLE 5. Link evolution, Hyderabad	12

5.4	TABLE 6. Link evolution for information and financial links and TABLE 7. Triples evolution, Karnataka	13
5.5	TABLE 8. Triples evolution, Hyderabad and TABLE 9. Borrowing patterns	14
5.6	TABLE 10. Risk sharing, Hyderabad	15
6	Short summary	15
7	Conclusion	16

1 Big picture

- **Question.** What happens to informal social networks when a formal credit market enters a community? The paper asks whether microfinance merely substitutes for some informal credit links or whether it changes the broader social network, including relationships among households that do not directly use microcredit.
- **Core empirical puzzle.** The surprising result is not only that microfinance reduces network links. The key puzzle is that links among households classified as unlikely to take up microfinance—the *L* types—fall at least as much as links involving likely borrowers, the *H* types.
- **Two empirical settings.** The paper uses two independent Indian settings:
 - **Karnataka:** 75 villages, two waves of detailed network data, and non-random but plausibly idiosyncratic entry by the MFI BSS into 43 villages.
 - **Hyderabad:** an RCT in 104 neighbourhoods, where the MFI Spandana initially entered 52 neighbourhoods and later entered the control group. Network outcomes are measured years later, after microfinance operations had shut down.
- **Main reduced-form fact.** In Karnataka, network density falls by about 0.012–0.013 relative to a non-MF mean of 0.113, roughly an 11% decline. In Hyderabad, treatment also lowers link density, with the preferred controlled estimate around -0.005 relative to a non-MF mean of 0.0318.
- **Main heterogeneity fact.** In Karnataka, an *LL* link that existed before microfinance is 5.8 percentage points less likely to survive in MF villages. New *LL* links are also less likely to form. In Hyderabad, treated neighbourhoods have about 0.7 percentage points fewer *LL* links. These estimates are not smaller than the effects on links involving *H* types.
- **Group-structure fact.** The same pattern appears for triangles. *LLL* triangles in Karnataka are more likely to dissolve in MF villages, and Hyderabad treatment also lowers the probability of observing *LLL* triangles.
- **Mechanism proposed by the paper.** Standard local network-formation models have difficulty explaining why relationships among non-borrowers decline so much. The authors propose a dynamic model of **undirected socializing effort**: people choose how much to go out and socialize, but whom they meet is partly random. When likely borrowers reduce socializing effort, everyone else faces a lower return to socializing, creating a global network-level externality.
- **Welfare-relevant margin.** The paper links network decline to informal borrowing and risk sharing. Households unlikely to take up microcredit lose informal borrowing capacity, and in Hyderabad their consumption becomes more sensitive to income fluctuations.
- **Takeaway.** Formal financial development can weaken informal networks in ways that hurt non-participants. The paper is therefore not simply about microfinance; it is about how market expansion can reshape social capital and generate spillovers outside the set of direct adopters.

2 Data and measurement (key objects)

2.1 Two contexts

Karnataka.

- The MFI is Bharatha Swamukti Samsthe (BSS).
- The study covers 75 villages in five districts of Karnataka.
- Baseline survey: 2006, before BSS entry.
- BSS entered 43 villages by 2010. Expansion then stopped during the Andhra Pradesh microcredit crisis, leaving 32 comparison villages without BSS exposure before the second survey.
- Wave 2 survey: 2012.
- The key strength is panel network data: the paper can study whether old links survive and whether new links form.

Hyderabad.

- The MFI is Spandana.
- In 2006, 52 of 104 neighbourhoods were randomly assigned to early entry.
- The remaining 52 neighbourhoods received access about two years later.
- In 2010, microfinance lending stopped because of the Andhra Pradesh crisis.
- In 2012, the survey measured networks. At that point, the early-entry group had had longer cumulative exposure to microfinance but not contemporaneous access.
- The key strength is random assignment; the key limitation is that the network data are cross-sectional and partially observed.

2.2 Network measurement

Karnataka network data.

- Adults report whom they interact with across 12 dimensions: visits, relatives and non-relatives they socialize with, medical help, money borrowing/lending, goods borrowing/lending, advice, and prayer.
- The main network is a household-level, undirected, unweighted network. A link exists if any member of one household is linked to any member of the other household on at least one dimension.
- The paper also separately studies financial links and informational links.
- Wave 1 covers network data from 46% of households; Wave 2 covers 89.14% of households. Because links are undirected, the authors observe a much larger share of potential links than the household sampling rate alone would suggest.

Hyderabad network data.

- The paper collects direct degree reports and Aggregated Relational Data (ARD).
- Direct degree questions ask how many neighbourhood links a household has along financial, social, and informational dimensions.
- ARD questions ask how many people the respondent knows with particular traits. The authors use the Breza et al. methodology to recover network statistics such as link probabilities, clustering, and triangle probabilities.

2.3 Classifying households as likely or unlikely microfinance borrowers

The paper divides households into two types:

- H : high predicted likelihood of joining microfinance if it is available.
- L : low predicted likelihood of joining microfinance.

The classifier is estimated using random forests on households exposed to microfinance, then applied to both MF and non-MF locations. The goal is to compare an H household in a control village with a similar household that would have been likely to borrow had microfinance been available.

Karnataka predictors. Eligibility through adult female household members, BSS leader status, closeness to leaders, closeness to same-caste leaders, and share of same-caste leaders.

Hyderabad predictors. Household and neighbourhood baseline characteristics, including household demographics, business ownership, and neighbourhood-level variables.

Interpretation. This classification is central because the paper’s strongest claim is about spillovers onto L households. If L households are truly unlikely to borrow, then losses in LL links and LLL triangles cannot be explained by direct microcredit participation.

3 Empirical specifications and model (all equations + interpretation)

3.1 Graph-level difference-in-differences in Karnataka

For Karnataka, the paper estimates effects on village-level network statistics using

$$y(g_{vt}) = \alpha + \beta \text{MF}_v \times \text{Post}_t + \gamma \text{MF}_v + \eta \text{Post}_t + \delta' X_v + \varepsilon_{vt}.$$

- g_{vt} is the network in village v and period t .
- $y(g_{vt})$ is a graph statistic: density, clustering, or closeness.
- MF_v indicates villages that BSS entered.
- Post_t indicates Wave 2.
- X_v includes baseline village/network controls; in richer specifications, controls are selected using double post-LASSO and village fixed effects are included.

Read. The coefficient β measures whether networks changed more in MF villages than in non-MF villages after BSS entry. The identifying assumption is parallel trends in network evolution, conditional on controls.

3.2 Cross-sectional treatment effect in Hyderabad

In Hyderabad, treatment assignment is randomized. For graph-level outcomes the key specification is a cross-sectional comparison of early-entry and later-entry neighbourhoods in 2012:

$$y(g_v) = \alpha + \beta \text{MF}_v + \delta' X_v + \varepsilon_v.$$

Read. Here MF_v means early exposure to Spandana. Because both treatment and control areas had eventually received access, and because lending had stopped by the 2012 network survey, the coefficient captures the persistent effect of longer cumulative exposure rather than contemporaneous borrowing.

3.3 Link-level evolution by household type

Let i and j be two households. In Karnataka, the authors condition either on baseline links that existed or on baseline non-links. A compact version of the link-level specification is

$$y_{ijv,2} = \alpha + \beta \text{MF}_v + \beta_{LH} \text{MF}_v \times LH_{ijv} + \beta_{HH} \text{MF}_v \times HH_{ijv} \\ + \gamma_{LH} LH_{ijv} + \gamma_{HH} HH_{ijv} + \delta' X_{ijv} + \varepsilon_{ijv,2}.$$

- $y_{ijv,2} = 1$ if the link exists in Wave 2.
- The omitted type is LL .
- LH_{ijv} and HH_{ijv} identify links involving one or two high-propensity households.
- X_{ijv} includes flexible controls for centrality, household demographics, and classifier variables, interacted with treatment where relevant.

Read. β is the treatment effect on LL links. If microfinance only directly affects borrowers, one would expect β to be small relative to the treatment effects on LH or HH links. Instead, LL links decline strongly.

3.4 Triangle-level evolution

For a triple i, j, k , the paper estimates

$$y_{ijkv,2} = \alpha + \beta \text{MF}_v + \beta_{LLH} \text{MF}_v \times LLH_{ijkv} + \beta_{LHH} \text{MF}_v \times LHH_{ijkv} \\ + \beta_{HHH} \text{MF}_v \times HHH_{ijkv} + \gamma_{LLH} LLH_{ijkv} + \gamma_{LHH} LHH_{ijkv} \\ + \gamma_{HHH} HHH_{ijkv} + \delta' X_{ijkv} + \varepsilon_{ijkv,2}.$$

The omitted category is LLL . In Karnataka, the authors can condition on triangles existing in Wave 1 and study whether they survive. In Hyderabad, they use ARD-estimated cross-sectional triangle probabilities.

Read. The coefficient on MF_v is the effect on LLL triangles. The key finding is that LLL triangles decline at least as much as other types of triangles, even though none of the three households are likely microfinance borrowers.

3.5 Risk-sharing specification

The paper studies whether consumption becomes more sensitive to income fluctuations in Hyderabad using

$$c_{ivt} = \alpha_i + \gamma_{vt} + \beta_1 y_{ivt} + \beta_2 y_{ivt} \times \text{MF}_v + \beta_3 y_{ivt} \times H_i + \beta_4 y_{ivt} \times \text{MF}_v \times H_i + \delta' X_{iv} + \varepsilon_{ivt}.$$

- c_{ivt} is per-capita consumption or expenditure.
- y_{ivt} is per-capita income.
- α_i are household fixed effects.
- γ_{vt} are neighbourhood-by-time fixed effects.
- β_2 tests whether low-propensity households in MF areas have more income pass-through into consumption.

Read. Under better risk sharing, idiosyncratic income changes should have less effect on consumption. A positive β_2 means L households exposed to microfinance have weaker consumption smoothing.

3.6 Network-formation model: undirected socializing effort

The model introduces a global externality through socializing effort.

There are n agents. Agent i has type θ_i . A type- θ agent gets base value $v_{\theta\theta'}$ from a link with a type- θ' agent, plus idiosyncratic match value ε_{ij} . The expected value of a mutually acceptable link is

$$E^+[v] \equiv \mathbb{E}[v + \varepsilon_{ij} \mid \varepsilon_{ij} > -v].$$

Agents choose socializing effort $e_i \in [0, 1]$. If agents i and j choose efforts e_i and e_j , they meet with probability proportional to $e_i e_j$.

Link maintenance. If i and j were linked last period, the link survives with probability $e_i e_j$.

Link formation. If i and j were not linked, the link forms with probability

$$e_i e_j \left(1 - F(-v_{\theta_i \theta_j})\right) \left(1 - F(-v_{\theta_j \theta_i})\right),$$

which requires both meeting and mutual consent.

Let

$$m_{\theta\theta'} = n_{\theta\theta'} (1 - F(-v_{\theta\theta'})) (1 - F(-v_{\theta'\theta})).$$

Then steady-state expected degree from type θ' friends is

$$d_{\theta\theta'} = m_{\theta\theta'} e_{\theta} e_{\theta'}.$$

The type- θ utility from effort is

$$V_{\theta}(e_{\theta}) = u_{\theta} e_{\theta} - \frac{1}{2} c_{\theta} e_{\theta}^2 + \sum_{\theta'} E^+[v_{\theta\theta'}] n_{\theta\theta'} (1 - F(-v_{\theta\theta'})) (1 - F(-v_{\theta'\theta})) e_{\theta'} e_{\theta}.$$

The interior best response is

$$e_\theta = \frac{1}{c_\theta} \left[u_\theta + \sum_{\theta'} E^+[v_{\theta\theta'}] n_{\theta\theta'} (1 - F(-v_{\theta\theta'})) (1 - F(-v_{\theta'\theta})) e_{\theta'} \right],$$

with the solution capped at one if needed.

Key proposition. For sufficiently high effort costs, the equilibrium is unique and stable. If the value of any relationship type falls, all types reduce effort, and all expected degrees fall. The model therefore generates global spillovers: even LL links can fall when only H -related link values change.

3.7 Specialization to microfinance

Let type θ have probability α_θ of having money to lend and probability β_θ of needing to borrow. Lending yields net return r ; borrowing from a friend yields net benefit b . The base payoff from a link is

$$v_{\theta\theta'} = \alpha_\theta \beta_{\theta'} r + \beta_\theta \alpha_{\theta'} b.$$

Microfinance changes the borrowing and lending needs of H types:

$$\Delta\beta_H = \beta'_H - \beta_H, \quad \Delta\alpha_H = \alpha'_H - \alpha_H.$$

The corresponding first-order changes in link values are approximately

$$\begin{aligned} \Delta v_{HH} &= (\alpha_H \Delta\beta_H + \beta_H \Delta\alpha_H)(r + b), \\ \Delta v_{LL} &= 0, \\ \Delta v_{HL} &= \alpha_L \Delta\beta_H b + \beta_L \Delta\alpha_H r, \\ \Delta v_{LH} &= \alpha_L \Delta\beta_H r + \beta_L \Delta\alpha_H b. \end{aligned}$$

Economic intuition. Even if $\Delta v_{LL} = 0$ directly, LL links can decline because socializing effort is a strategic complement. When H types become less valuable as relationship partners, L types may reduce undirected socializing effort, which lowers the meeting rate between L households themselves.

4 Identification and instruments (what makes the estimates credible)

4.1 Karnataka: difference-in-differences

The Karnataka design is not randomized. Its credibility comes from the timing of BSS expansion and the panel structure of the network data.

- BSS planned to enter all 75 villages, but expansion stopped because of the Andhra Pradesh microcredit crisis.
- The paper can compare network changes from Wave 1 to Wave 2 in MF versus non-MF villages.
- Rich baseline controls and double post-LASSO address concerns that BSS entered villages with different baseline characteristics.

- Because the authors observe Wave 1 links, they can distinguish link survival from new link formation.

Main limitation. Since entry was not randomly assigned, parallel trends cannot be proven. Village size and other pre-treatment differences are important concerns. The paper responds by controlling flexibly for baseline village and network characteristics.

4.2 Hyderabad: randomized early entry

The Hyderabad design is stronger causally because initial access was randomized.

- Early entry was randomly assigned across 104 neighbourhoods.
- Later-entry neighbourhoods provide a comparison group.
- By the 2012 network survey, both groups had lost access to microfinance, so the estimates capture durable effects of differential exposure.

Main limitation. The network data are only measured at endline and are partially observed. The paper relies on ARD to estimate unobserved network statistics.

4.3 Why the two settings complement each other

- Karnataka has high-quality panel network data but non-random treatment.
- Hyderabad has randomized treatment but weaker network measurement.
- The same qualitative pattern appears in both settings.

Credibility takeaway. The most persuasive evidence is not a single estimate; it is the replication of the same unexpected *LL* and *LLL* spillover pattern across two very different empirical designs.

5 Tables and extracted visuals (original numbers and titles)

5.1 TABLE 1. Sample statistics and TABLE 1. Continued

TABLE 1. Sample statistics

	Obs	Control mean	Control SD	Treatment—control		
				Coeff.	5% limit	p-Value
<i>Panel A: Karnataka Wave 1 Households</i>						
Eligible Female	7511	0.943	0.233	0.008	0.015	0.216
No access to latrine	7511	0.748	0.434	−0.038	0.051	0.205
Number of rooms	7511	2.489	1.313	−0.001	0.140	0.973
Thatched roof	7511	0.021	0.145	−0.002	0.014	0.678
Distance to Bangalore	7511	61.114	17.458	−3.823	8.074	0.309
All loans	7218	37,861.564	129,797.423	1351.740	11,597.294	0.819
Network (friends and family) loans	7218	2735.470	25,394.731	6.467	1716.401	0.994
SHG loans	7218	2543.994	6944.324	14.783	968.668	0.976
Bank loans	7218	19,892.356	106,358.225	3563.106	8808.589	0.428
Moneylender loans	7218	3638.339	20,456.671	−164.660	1656.949	0.846
Distance to leaders > 1	7511	0.206	0.404	0.042	0.056	0.163
Leader	7511	0.154	0.361	−0.007	0.021	0.555
Share WC leader	7511	0.516	0.294	−0.060	0.078	0.112
No electricity	7511	0.075	0.263	−0.018	0.022	0.134
GMOBC	7511	0.709	0.454	−0.039	0.065	0.221
Num. beds	7511	0.912	1.222	0.022	0.213	0.737
RCC roof	7511	0.147	0.355	−0.031	0.045	0.167
Household size	7511	5.014	2.205	0.297	0.231	0.013
Own rent	7511	0.100	0.300	−0.009	0.044	0.810
Distance to town	7511	5.647	3.595	1.203	1.862	0.190
H (RF)	7511	0.542	0.498	−0.209	0.068	0.000
H (logit)	7511	0.752	0.432	0.005	0.042	0.800
<i>Panel B: Hyderabad</i>						
Total outstanding debt in area, baseline	104	39675.337	47776.778	−6981.245	13853.634	0.326
Area population, baseline	104	264.615	160.467	−3.385	58.378	0.910
Total number of businesses in area, baseline	104	7.288	5.003	−0.346	1.927	0.726
Area mean monthly per-capita exp, baseline	104	1004.974	171.510	42.847	70.733	0.238
Area literacy rate (HH heads), baseline	104	0.625	0.167	0.007	0.056	0.811
Area literacy rate (all), baseline	104	0.687	0.094	0.000	0.032	0.976
Prime-aged (18–45) women in HH, endline 1	6863	1.456	0.820	−0.024	0.056	0.413
Owens land in Hyderabad, endline 1	6863	0.061	0.239	−0.001	0.014	0.897
Owens land in village, endline 1	6863	0.194	0.396	0.006	0.057	0.841
HH had a business pre-intervention	6863	0.308	0.462	0.007	0.042	0.736
HH size (adult equiv), endline 1	6863	4.690	1.784	−0.008	0.130	0.899
Adults (16+) in HH, endline 1	6863	3.887	1.754	−0.018	0.129	0.780
Children (< 15) in HH, endline 1	6863	1.738	1.310	−0.014	0.104	0.797

(continued)

TABLE 1. Continued

	Obs	Control mean	Control SD	Treatment—control		
				Coeff.	5% limit	p-Value
Male head of household, endline 1	6863	0.895	0.307	0.012	0.021	0.266
Age of head of household, endline 1	6863	41.146	10.228	−0.226	0.774	0.566
Head of HH with no education, endline 1	6863	0.312	0.463	0.001	0.044	0.975
Any child 13–18 in HH, endline 1	6863	0.452	0.498	0.016	0.031	0.305
Spouse is literate, endline 1	6863	0.543	0.498	0.003	0.049	0.919
Spouse works for a wage, endline 1	6863	0.234	0.423	−0.020	0.048	0.405
H (RF)	6863	0.241	0.427	−0.016	0.077	0.684
H (logit)	6863	0.236	0.425	0.010	0.050	0.705

Notes: This table presents summary statistics and baseline balance for Karnataka (Panel A) and Hyderabad (Panel B). The “5% limit” column shows how large the difference between treatment and control would have needed to be significant at the 5% level. The “p-value” column shows a test of significance on the difference between treatment and control. In Karnataka, there are 75 villages in the sample; 43 received MF. In Hyderabad, 104 neighbourhoods were subject to randomized assignment of MF. GMOBC = A dummy for whether the household is general caste or other backwards caste; the omitted categories are scheduled caste and scheduled tribes. General and OBC are considered upper caste. RCC is Reinforced Cement Concrete. The values labeled “baseline” in Hyderabad are at the neighbourhood level, while “endline 1” values are at the household level. $H(RF)$ is the random forest classification into high ($H = 1$) or low ($H = 0$) MF propensity. $H(logit)$ is the corresponding logit classification: see Section 2.4 for details.

Read. Table 1 documents balance and descriptive statistics. Karnataka has one statistically significant household-level imbalance out of many variables, but the random-forest H classification is imbalanced in Karnataka. Hyderabad, where initial access was randomized, is balanced across baseline neighbourhood and household characteristics.

Why it matters. The table motivates the paper’s different identification strategies: Karnataka requires careful controls; Hyderabad relies on random assignment.

5.2 TABLE 2. Characteristics of H versus L and TABLE 3. Effects of MF on graph-level characteristics

TABLE 2. Characteristics of H versus L

TABLE 2
Characteristics of H versus L

Panel A: Karnataka—Demographics and Amenities Variables

	(1) GMOBC	(2) Latrine	(3) Num. Rooms	(4) Num. Beds	(5) Thatched Roof	(6) RCC Roof
H	-0.221 (0.021) [0.000]	-0.115 (0.018) [0.000]	-0.177 (0.042) [0.000]	-0.213 (0.038) [0.000]	0.019 (0.005) [0.000]	-0.053 (0.008) [0.000]
Depvar Mean	0.7	0.261	2.36	0.84	0.0235	0.117
Observations	14,904	14,904	14,904	14,904	14,904	14,904

Panel B: Karnataka—Network Variables

	(1) Degree	(2) Links to L	(3) Links to H	(4) Eig. Cent.
H	1.947 (0.254) [0.000]	-0.432 (0.208) [0.000]	2.355 (0.164) [0.009]	0.017 (0.002) [0.000]
Depvar Mean	8.97	4.61	3.09	0.0524
Observations	14,904	14,904	14,904	14,904

Panel C: Hyderabad—Demographics and Amenities Variables

	(1) GMOBC	(2) Latrine	(3) Num. Rooms	(4) Thatched Roof	(5) RCC Roof
H	0.007 (0.040) [0.852]	0.041 (0.030) [0.167]	0.185 (0.104) [0.076]	0.001 (0.010) [0.954]	-0.026 (0.027) [0.338]
Observations	4520	4483	4516	4516	4508
Depvar Mean	0.429	0.578	2.314	0.025	0.882

Panel D: Hyderabad—Network Variables

	(1) Exp. Degree	(2) Exp. Links to L	(3) Exp. Links to H	(4) Exp. Eig. Cent.
H	0.182 (0.149) [0.222]	-0.651 (0.194) [0.001]	0.834 (0.215) [0.000]	0.009 (0.005) [0.074]
Observations	4523	4523	4523	4523
Depvar Mean	5.806	4.379	1.428	0.074

Notes: Standard errors (clustered at the village level) are reported in parentheses. p -values are reported in brackets. Panels A and B pertain to Karnataka, based on Wave 1 data only. In Panels C and D pertaining to Hyderabad, the estimates reflect H -vs- L differences for the non-MF (control group) sample only. GMOBC = A dummy for whether the household consists of general caste or other backwards caste, so the omitted categories are scheduled caste and scheduled tribes. General and OBC are considered upper caste. RCC is Reinforced Cement Concrete.

TABLE 3. Effects of MF on graph-level characteristics

TABLE 3
Effects of MF on graph-level characteristics

Panel A: Karnataka

	(1) Density	(2) Density	(3) Density	(4) Clustering	(5) Clustering	(6) Clustering	(7) Closeness	(8) Closeness	(9) Closeness
MF $\times$ Post	-0.012 (0.007) [0.004]	-0.013 (0.007) [0.007]	-0.013 (0.007) [0.007]	0.004 (0.015) [0.807]	0.010 (0.007) [0.083]	0.010 (0.016) [0.533]	-0.022 (0.010) [0.024]	-0.018 (0.012) [0.128]	-0.016 (0.012) [0.188]
Microfinance	-0.020 (0.008) [0.018]	0.006 (0.005) [0.277]	0.007 (0.006) [0.235]	-0.041 (0.016) [0.013]	-0.018 (0.006) [0.235]	-0.013 (0.015) [0.377]	-0.013 (0.010) [0.199]	0.011 (0.010) [0.239]	0.012 (0.011) [0.258]
Post	-0.012 (0.006) [0.046]	-0.014 (0.011) [0.183]	-0.014 (0.011) [0.206]	-0.009 (0.010) [0.366]	0.009 (0.011) [0.206]	0.009 (0.026) [0.745]	0.105 (0.008) [0.000]	0.064 (0.057) [0.266]	-0.042 (0.055) [0.448]
Observations	150	150	150	150	150	150	150	150	150
Double-Post LASSO		$\checkmark$	$\checkmark$		$\checkmark$	$\checkmark$		$\checkmark$	$\checkmark$
Village FE			$\checkmark$		$\checkmark$	$\checkmark$		$\checkmark$	$\checkmark$
Non-MF Mean	0.113	0.113	0.113	0.329	0.329	0.329	0.431	0.431	0.431
Depvar Mean	0.0983	0.0983	0.0983	0.307	0.307	0.307	0.418	0.418	0.418

Panel B: Hyderabad

	(1) Density	(2) Density	(3) Clustering	(4) Clustering	(5) Closeness	(6) Closeness
MF	-0.006 (0.003) [0.086]	-0.005 (0.003) [0.062]	-0.006 (0.007) [0.426]	-0.002 (0.007) [0.771]	-0.002 (0.002) [0.501]	-0.001 (0.002) [0.575]
Double-Post LASSO	No	Yes	No	Yes	No	Yes
Depvar Mean	0.0286	0.0286	0.052	0.052	0.00262	0.00262
Non-MF Mean	0.0318	0.0318	0.053	0.053	0.00312	0.00312
Observations	4429	4429	89	89	89	89

Notes: Standard errors (clustered at the village level) are reported in parentheses. p -values are reported in brackets. For Panel A (Karnataka), controls consist of the share of upper caste households, number of households in the village, share of households in self-help groups, share Hindus, share with a latrine in the house, share that own the household, share that have electricity and share that are leaders. For Panel B (Hyderabad), controls are the same demographic characteristics of households and villages that are used in random forest classification of H versus L . In Panel B, Columns (1) and (2) for density were found through direct elicitation at the node level, whereas columns (3)–(6) for clustering and closeness were estimated at the graph level through the ARD survey.

Read Table 2. H households are poorer and more central in Karnataka. In Hyderabad, H households are not sharply different demographically in control areas, but they have different network positions: fewer expected links to L types, more expected links to H types, and greater centrality.

Read Table 3. Microfinance reduces overall network density in both Karnataka and Hyderabad. In Karnataka, the MF $\times$ Post coefficient for density is around -0.012 to -0.013 . In Hyderabad, the controlled density estimate is -0.005 . Clustering and closeness effects are less stable.

Why it matters. Table 3 establishes the aggregate fact: formal credit exposure shrinks the social network. The rest of the paper asks where in the network this shrinkage occurs.

5.3 TABLE 4. Link evolution, Karnataka and TABLE 5. Link evolution, Hyderabad

TABLE 4. Link evolution, Karnataka

TABLE 4 Link evolution, Karnataka				
	(1) Linked Post-MF	(2) Linked Post-MF	(3) Linked Post-MF	(4) Linked Post-MF
MF	-0.058 (0.018) [0.002]	-0.059 (0.019) [0.002]	-0.023 (0.008) [0.006]	-0.021 (0.008) [0.008]
MF $\times$ LH	0.009 (0.015) [0.573]	0.001 (0.014) [0.935]	0.007 (0.004) [0.120]	0.007 (0.004) [0.109]
MF $\times$ HH	0.039 (0.022) [0.086]	0.023 (0.022) [0.292]	0.009 (0.007) [0.206]	0.012 (0.007) [0.059]
LH	-0.025 (0.012) [0.036]	-0.005 (0.011) [0.637]	-0.002 (0.004) [0.566]	-0.006 (0.004) [0.095]
HH	0.008 (0.017) [0.622]	0.041 (0.017) [0.020]	0.021 (0.006) [0.001]	0.008 (0.006) [0.189]
Observations	57,376	57,376	846,561	846,561
Linked Pre-MF	Yes	Yes	No	No
Controls	✓	✓	✓	✓
Depvar Mean	0.441	0.441	0.0636	0.0636
LL, Non-MF Mean	0.482	0.482	0.0753	0.0753
MF + MF $\times$ LH = 0 p -val	0.014	0.005	0.015	0.014
MF + MF $\times$ HH = 0 p -val	0.361	0.088	0.101	0.232
MF + LH $\times$ MF = MF + HH $\times$ MF p -val	0.137	0.286	0.641	0.245

Notes: Standard errors (clustered at the village level) are reported in parentheses, p -values are reported in brackets. Controls are selected by double post lasso among centrality controls (vector of flexible controls for centrality of both nodes), household characteristics (caste, a number of wealth proxies including number of rooms, number of beds, electrification, latrine presence, and roofing material) and all variables that are used in the random forest classification.

TABLE 5. Link evolution, Hyderabad

TABLE 5 Link evolution, Hyderabad		
	(1) Prob. Linked	(2) Prob. Linked
MF	-0.005 (0.002) [0.035]	-0.007 (0.002) [0.004]
MF $\times$ LH	0.002 (0.003) [0.577]	-0.001 (0.002) [0.764]
MF $\times$ HH	-0.011 (0.008) [0.203]	-0.007 (0.006) [0.281]
LH	0.002 (0.003) [0.532]	0.003 (0.002) [0.113]
HH	0.018 (0.008) [0.020]	0.014 (0.006) [0.024]
Observations	141,996	141,996
Controls	No	Yes
Depvar Mean	0.0255	0.0255
LL, Non-MF Mean	0.0268	0.0268
MF + MF $\times$ LH = 0 p -val	0.387	0.019
MF + MF $\times$ HH = 0 p -val	0.066	0.041
MF + MF $\times$ HH = MF + MF $\times$ LH p -val	0.038	0.18

Notes: Standard errors (clustered at the village level) are reported in parentheses, p -values are reported in brackets. The controls are selected by double post lasso among all variables that are used for random forest classification.

Read Table 4. In Karnataka, baseline LL links are 5.8–5.9 percentage points less likely to survive in MF villages. New LL links are also 2.1–2.3 percentage points less likely to form. The effects on LH and HH links are not larger, which is the central spillover finding.

Read Table 5. In Hyderabad, treated neighbourhoods have about 0.5–0.7 percentage points fewer LL links. Again, the reductions are not concentrated only among links involving H types.

Why it matters. These tables show that non-borrower relationships are directly affected by microfinance entry, even though neither party is likely to borrow.

5.4 TABLE 6. Link evolution for information and financial links and TABLE 7. Triples evolution, Karnataka

TABLE 6. Link evolution for information and financial links

Panel A: Karnataka				
	(1) Financial Linked Post-MF	(2) Financial Linked Post-MF	(3) Info Linked Post-MF	(4) Info Linked Post-MF
MF	-0.052 (0.021) [0.014]	-0.012 (0.005) [0.016]	-0.050 (0.018) [0.006]	-0.015 (0.005) [0.007]
MF $\times$ LH	-0.005 (0.019) [0.813]	0.003 (0.003) [0.213]	0.002 (0.017) [0.892]	0.006 (0.003) [0.076]
MF $\times$ HH	0.029 (0.026) [0.263]	0.004 (0.005) [0.373]	0.040 (0.021) [0.063]	0.005 (0.005) [0.284]
LH	-0.015 (0.015) [0.331]	-0.001 (0.003) [0.623]	-0.014 (0.013) [0.288]	-0.004 (0.003) [0.183]
HH	-0.00002 (0.022) [1.000]	0.013 (0.004) [0.003]	-0.004 (0.017) [0.813]	0.012 (0.004) [0.005]
Observations	27,072	876,865	37,044	866,893
Linked Pre-MF	Yes	No	Yes	No
Depvar Mean	0.333	0.0341	0.326	0.0377
LL, Non-MF Mean	0.371	0.04	0.361	0.0464
MF + MF $\times$ LH = 0 p -val	0.005	0.035	0.008	0.025
MF + MF $\times$ HH = 0 p -val	0.343	0.157	0.615	0.061
MF + LH $\times$ MF = MF + HH $\times$ MF p -val	0.14	0.828	0.046	0.898

	(1) Number of Financial Links (Degree)	(2) Number of Financial Links (Degree)	(3) Number of Non Financial Links (Degree)	(4) Number of Non Financial Links (Degree)
MF	-0.363 (0.137) [0.010]	-0.393 (0.140) [0.006]	-0.195 (0.118) [0.101]	-0.157 (0.113) [0.169]
MF $\times$ H	0.571 (0.214) [0.010]	0.748 (0.203) [0.0004]	0.651 (0.212) [0.003]	0.675 (0.206) [0.002]
H	0.071 (0.149) [0.634]	-0.195 (0.145) [0.182]	-0.191 (0.164) [0.247]	-0.298 (0.165) [0.074]
Observations	4429	4429	4429	4429
Double-Post LASSO	No	No	No	Yes
Depvar Mean	4.24	4.24	2.87	2.87
MF + MFxH = 0 p -val	0.362	0.090	0.045	0.013

Notes: Standard errors (clustered at the village level) are reported in parentheses. p -values are reported in brackets.
Panel A: Columns 1–2 restrict to financial links; columns 3–4 restrict to non-financial links. Columns 1 and 3 consider links that existed in Wave 1, while columns 2 and 4 consider pairs of nodes that were not linked in Wave 1.
Panel B: Dependent variables in all columns are node-level self-reported counts of financial versus non-financial links. All columns include a full set of controls. Centrality controls are a vector of flexible controls (a polynomial) for centrality of both nodes. Household characteristics are caste and wealth proxies including number of rooms, number of beds, electrification, latrine presence, and roofing material. Household predictor variables consist of all variables that are used in the random forest classification. In every case we include interactions of all of these network, demographic, and classification variables with the MF indicator.

TABLE 7. Triples evolution, Karnataka

	(1) Full triangle linked Post-MF	(2) Full triangle linked Post-MF	(3) Any link in triangle survived Post-MF	(4) Any link in triangle survived Post-MF
MF	-0.078 (0.029) [0.008]	-0.070 (0.026) [0.008]	-0.085 (0.023) [0.000]	-0.076 (0.019) [0.000]
MF $\times$ LLH	0.026 (0.021) [0.228]	0.015 (0.019) [0.437]	0.043 (0.018) [0.015]	0.029 (0.015) [0.050]
MF $\times$ LHH	0.054 (0.030) [0.072]	0.028 (0.025) [0.256]	0.057 (0.025) [0.022]	0.031 (0.018) [0.092]
MF $\times$ HHH	0.093 (0.042) [0.028]	0.049 (0.038) [0.199]	0.087 (0.031) [0.006]	0.048 (0.026) [0.061]
LLH	-0.024 (0.018) [0.180]	-0.003 (0.017) [0.879]	-0.037 (0.014) [0.009]	-0.019 (0.013) [0.137]
LHH	-0.037 (0.025) [0.133]	0.009 (0.023) [0.696]	-0.032 (0.017) [0.053]	0.002 (0.014) [0.871]
HHH	-0.025 (0.033) [0.454]	0.042 (0.029) [0.151]	-0.012 (0.022) [0.593]	0.034 (0.020) [0.090]
Observations	53,233	53,233	53,233	53,233
Linked Pre-MF	Yes	Yes	Yes	Yes
Controls		✓		✓
Depvar Mean	0.197	0.197	0.808	0.808
LL, Non-MF Mean	0.252	0.252	0.864	0.864
MF + MF $\times$ HHH = 0 p -val	0.698	0.549	0.935	0.209
MF + MF $\times$ LLH = 0 p -val	0.023	0.03	0.022	0.025
MF + MF $\times$ LHH = 0 p -val	0.262	0.048	0.141	0.018
MF + MF $\times$ HHH = MF + MF $\times$ LLH p -val	0.076	0.35	0.093	0.459
MF + MF $\times$ HHH = MF + MF $\times$ LHH p -val	0.212	0.492	0.075	0.307
MF + MF $\times$ LLH = MF + MF $\times$ LHH p -val	0.122	0.456	0.409	0.934

Notes: Standard errors (clustered at the village level) are reported in parentheses. p -values are reported in brackets. Controls are selected by double post lasso among centrality controls (vector of flexible controls for centrality of both nodes), household characteristics (caste, a number of wealth proxies including number of rooms, number of beds, electrification, latrine presence, and roofing material) and all variables that are used in the random forest classification.

Read Table 6. The decline is not restricted to financial links. In Karnataka, both financial and informational links fall. In Hyderabad, L households report fewer financial links, while H households are partly insulated or offset the reduction.

Read Table 7. In Karnataka, LLL triangles are especially vulnerable. They are 7.8 percentage points less likely to remain full triangles in MF villages, and at least one link in the triangle is less likely to survive.

Why it matters. Informal insurance often relies on supported links and local group structure. The triangle results show that microfinance affects not just pairwise ties but also the social architecture that helps sustain cooperation.

5.5 TABLE 8. Triples evolution, Hyderabad and TABLE 9. Borrowing patterns

TABLE 8. Triples evolution, Hyderabad

TABLE 8 Triples evolution, Hyderabad		
All variables $\times$ 1000	Full Triangle Linked (1)	Full Triangle Linked (2)
MF	-0.018 (0.010) [0.067]	-0.034 (0.020) [0.086]
MF $\times$ LLH	0.010 (0.011) [0.370]	-0.012 (0.013) [0.344]
MF $\times$ LHH	-0.027 (0.038) [0.472]	-0.052 (0.040) [0.191]
MF $\times$ HHH	-0.177 (0.097) [0.067]	-0.132 (0.089) [0.139]
LLH	0.000 (0.010) [0.976]	0.018 (0.013) [0.168]
LHH	0.055 (0.036) [0.132]	0.072 (0.042) [0.087]
HHH	0.217 (0.095) [0.023]	0.168 (0.093) [0.071]
Observations	3,341,006	3,341,006
Controls	No	Yes
Depvar Mean	0.0353	0.0353
LLL, Non-MF Mean	0.0359	0.0359
MF + MF $\times$ HHH = 0 p -val	0.045	0.087
MF + MF $\times$ LLH = 0 p -val	0.552	0.064
MF + MF $\times$ LHH = 0 p -val	0.256	0.072
MF + MF $\times$ HHH = MF + MF $\times$ LLH p -val	0.046	0.144
MF + MF $\times$ HHH = MF + MF $\times$ LHH p -val	0.041	0.162
MF + MF $\times$ LLH = MF + MF $\times$ LHH p -val	0.217	0.178

Notes: Standard errors (clustered at the village level) are reported in parentheses. p -values are reported in brackets. The controls are selected by double post lasso among all the variables that are used for its random forest classification, and includes several household and village level characteristics.

TABLE 9. Borrowing patterns

TABLE 9 Borrowing patterns					
	(1) MF	(2) Friends	(3) SHG	(4) Moneylender	(5) Family
Panel A: Borrowing Patterns, Karnataka					
MF $\times$ Post	476.572 (148.808) [0.002]	-562.308 (330.341) [0.089]	-844.524 (384.839) [0.029]	704.391 (800.168) [0.379]	677.970 (659.590) [0.305]
MF $\times$ Post $\times$ H	1795.233 (245.414) [0.000]	203.926 (242.383) [0.401]	48.466 (346.884) [0.889]	-2210.964 (943.562) [0.020]	-1608.814 (1185.489) [0.175]
MF $\times$ H	-0.542 (58.782) [0.993]	-65.457 (63.966) [0.307]	232.443 (254.356) [0.361]	206.495 (497.672) [0.679]	1088.834 (885.048) [0.219]
Post $\times$ H	189.508 (108.311) [0.081]	-410.031 (199.678) [0.041]	91.263 (279.643) [0.745]	1828.811 (734.643) [0.013]	400.044 (522.679) [0.445]
Observations	28,062	27,194	28,062	28,062	28,062
Depvar Mean	596.976	860.228	1863.324	2667.56	1656.881
L, Non-MF Mean	189.671	1148.705	1920.918	2344.905	1711.001
MF $\times$ Post $\times$ H + MF $\times$ Post = 0 p -val	0.000	0.255	0.119	0.084	0.325
Panel B: Borrowing Patterns, Hyderabad					
MF	-209.748 (235.127) [0.375]	86.742 (894.331) [0.923]	-1882.840 (801.110) [0.021]	-2664.192 (1455.603) [0.071]	-256.318 (656.431) [0.697]
MF $\times$ H	8312.670 (448.982) [0.000]	-637.232 (1491.449) [0.671]	-1577.128 (1369.064) [0.252]	4689.554 (2622.331) [0.077]	1796.860 (1366.622) [0.192]
H	-108.232 (296.017) [0.716]	-1792.590 (1293.944) [0.169]	1251.211 (1163.829) [0.285]	-198.590 (1899.306) [0.917]	-507.290 (985.862) [0.608]
Observations	6811	6863	6863	6863	6863
Depvar Mean	3107.86	7895.05	6935.66	18805.06	2620.97
L, Non MF Mean	2091.75	8110.94	7064.44	19601.47	2704.03
MF + MF $\times$ H = 0 p -val	0.000	0.664	0.012	0.426	0.245

Notes: Table presents the effect of MF access on the loan amounts borrowed from various sources; outcomes are winsorized at the 2.5% level. Panel A pertains to Karnataka and tracks loan amounts from MF institutions, friends, family, banks and moneylenders. All of its columns control for surveyed in wave 1 fixed effects. The average rate of inflation over the period between waves in Karnataka was 8% meaning a total of 65%. Panel B pertains to Hyderabad. Outcomes are measured in the first survey wave (2007–08). Here all specifications include demographic household and village controls (those used in random forest classification of H versus L) subject to double-post LASSO. Standard errors (clustered at the village level) are reported in parentheses. p -values are reported in brackets. MF: Microfinance Institution; SHG: Self-Help Group.

Read Table 8. Hyderabad also shows a reduction in LLL triangles, although the estimates are noisier because triangle probabilities are sparse and measured using ARD.

Read Table 9. In Karnataka, L households in MF villages borrow less from friends. In Hyderabad, L households reduce borrowing from self-help groups and moneylenders. H households, by contrast, gain access to microfinance loans.

Why it matters. The network changes are not purely cosmetic. They are connected to the loss of informal financial relationships, especially for households not directly served by microfinance.

5.6 TABLE 10. Risk sharing, Hyderabad

TABLE 10. Risk sharing, Hyderabad

TABLE 10 <i>Risk sharing, Hyderabad</i>		
	(1) Expenditures: Non-Food	(2) Expenditures: Total
MF $\times$ Income	0.071 (0.030) [0.022]	0.066 (0.037) [0.079]
MF $\times$ Income $\times H$	-0.065 (0.044) [0.153]	-0.112 (0.058) [0.070]
Household Income per capita	0.058 (0.019) [0.004]	0.109 (0.024) [0.000]
Household Income per capita $\times H$	0.020 (0.025) [0.438]	0.076 (0.043) [0.082]
Observations	10,502	10,590
Depvar Mean	1193	2040
L , Non-MF Depvar Mean	1187	2049
Income Mean	1440	1437
L , Non-MF Income Mean	1437	1435
Test: MF $\times$ Income + MF $\times$ Income $\times H = 0$	0.834	0.407

Notes: Income is total household, monthly per capita earnings from employment or business activities, excluding private and government transfers. Dependent variable is monthly per capita household expenditure. In col. 1, expenditure excludes food and in col. 2, we present non-food expenditure. Data is from the first (2007–08) and third (2012) waves of the Hyderabad survey. Regression includes controls for household fixed effects and wave-by-neighbourhood-by-type fixed effects. Additional controls are selected by double post lasso from the set of variables used in the prediction exercise, interacted with type. Standard errors (clustered at the neighbourhood level) are reported in parentheses. p -values are reported in brackets.

Read. In control areas, income pass-through into non-food consumption for L households is 0.058. In MF areas, that pass-through rises by 0.071, meaning L households become less insured against income fluctuations. For H households, the additional pass-through is offset by the interaction term, so they do not experience the same worsening of risk sharing.

Why it matters. Table 10 closes the loop between networks and welfare. The households least likely to borrow from microfinance appear to lose informal insurance when the community network weakens.

6 Short summary

- **Method.** The paper combines a Karnataka difference-in-differences design with panel network data and a Hyderabad RCT with ARD-estimated network statistics.
- **Classification.** Random forests classify households into likely microfinance borrowers (H) and unlikely borrowers (L).
- **Main empirical result.** Formal credit exposure reduces network density and weakens LL links and LLL triangles, even though L households are unlikely to take microfinance loans.
- **Main theoretical result.** A model of undirected socializing effort can generate global spillovers: when some households reduce the value of socializing, all households may reduce effort, and even unaffected dyads can lose links.
- **Welfare result.** L households lose informal borrowing opportunities and, in Hyderabad, become less able to smooth income shocks.

- **Policy lesson.** Formal financial access can have unintended distributional effects through social networks. Evaluating market expansion requires looking beyond direct adopters.

7 Conclusion

Core conclusion. Microfinance entry changes more than credit access. It changes the equilibrium structure of social networks, and the costs can fall heavily on households that do not participate in microfinance.

Economic intuition. Informal networks are maintained by costly social effort. Once formal credit changes the value of maintaining certain relationships, effort choices adjust at the community level. Because meetings are partly undirected, lower effort by one group reduces the return to effort for others, producing global spillovers.

Takeaway. The paper's contribution is to show that formal-market expansion can crowd out informal social capital in a non-local way. The most important beneficiaries of a formal market are not necessarily the only people affected by it, and some non-beneficiaries may be made worse off through weakened informal insurance and credit networks.